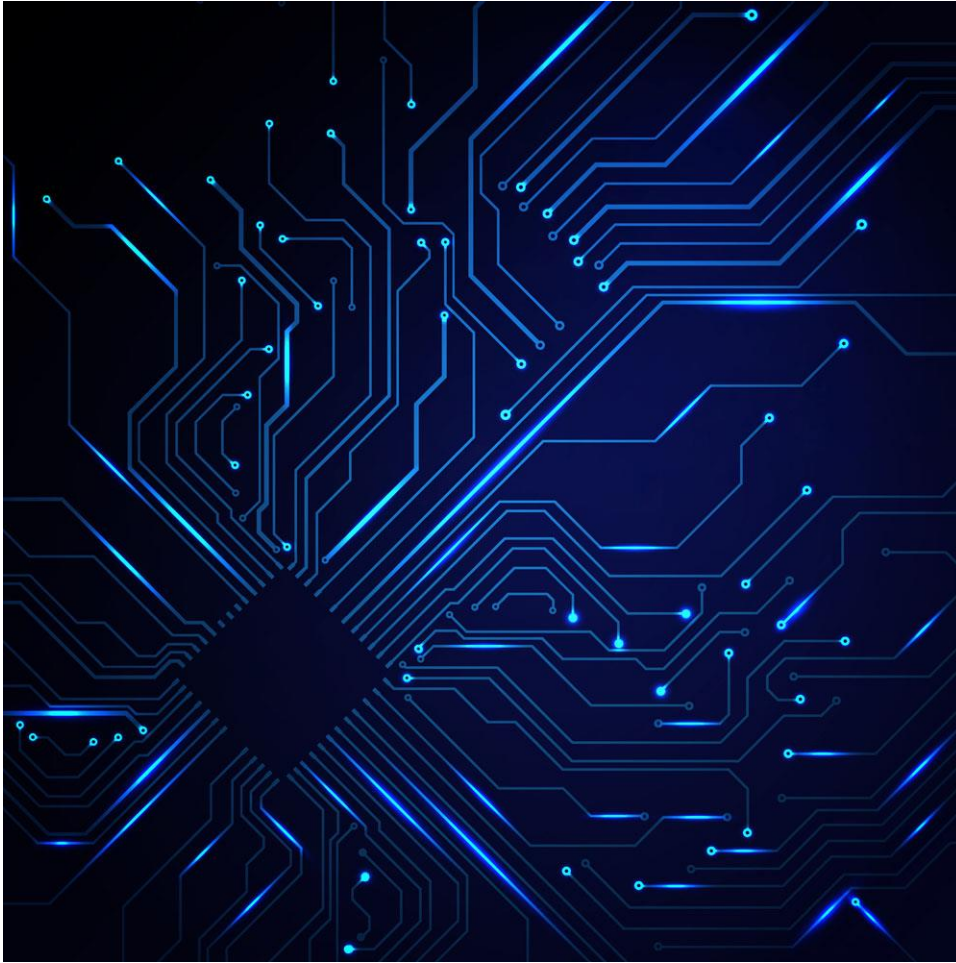




LAB 3 BASIC LOGIC GATE IMPLEMENTATION: PART 3

Group : 14

Name : 黃翊宏

Student ID : 01357101

Note

1. All files must be converted to PDF format before submission.
2. **Submission deadline:**
Next Tuesday at 1:00 PM
3. Each person submits one assignment.

☆ Lab 3-1

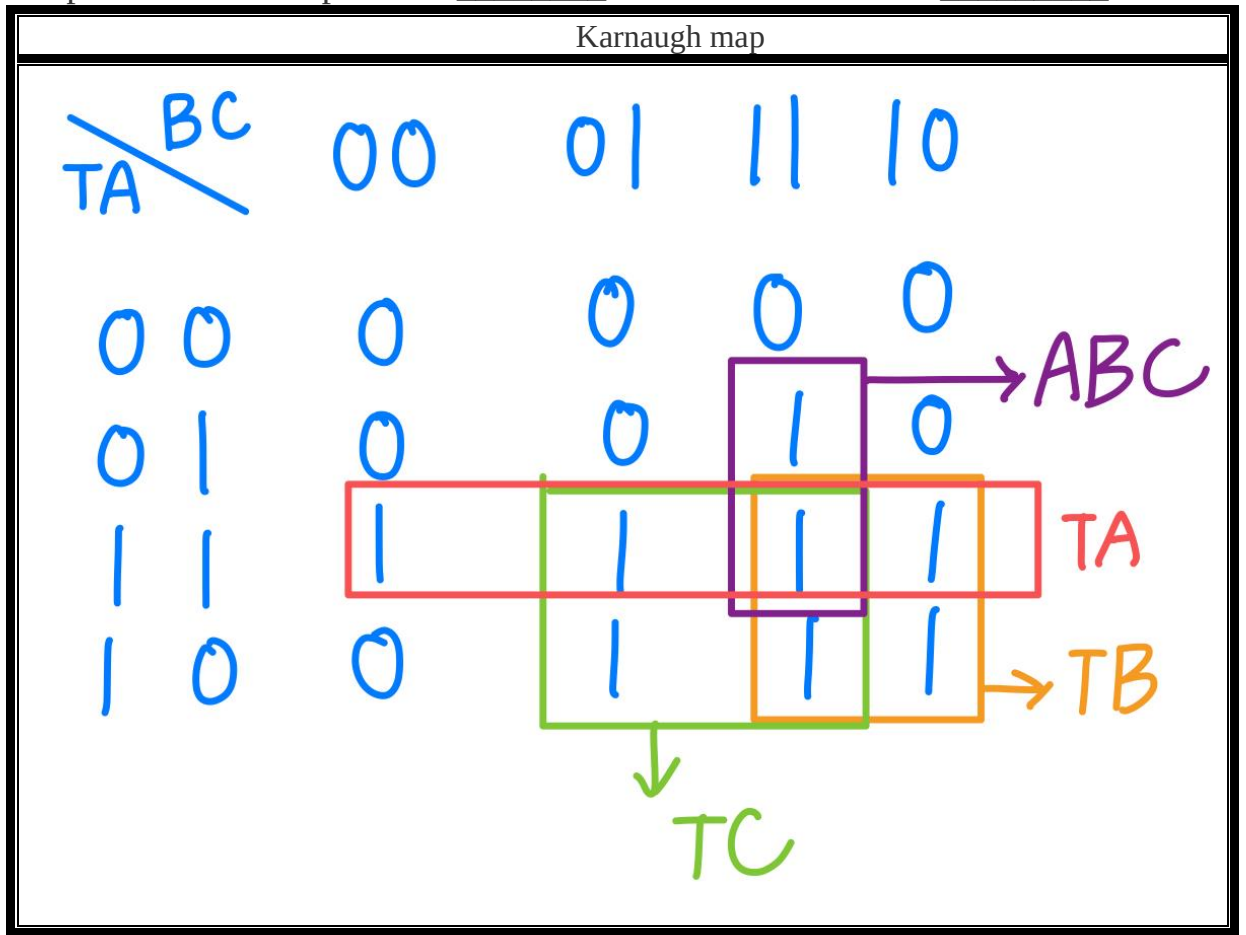
Truth Table (Logical)				
<i>T</i>	<i>A</i>	<i>B</i>	<i>C</i>	Out
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Truth Table (Voltage)				
T	A	B	C	Out
0	0	0	0	0.11
0	0	0	1	0.11
0	0	1	0	0.11
0	0	1	1	0.11
0	1	0	0	0.11
0	1	0	1	0.11
0	1	1	0	0.11
0	1	1	1	2.26
1	0	0	0	0.11
1	0	0	1	2.26
1	0	1	0	2.26
1	0	1	1	2.26
1	1	0	0	2.26
1	1	0	1	2.26
1	1	1	0	2.26
1	1	1	1	2.26



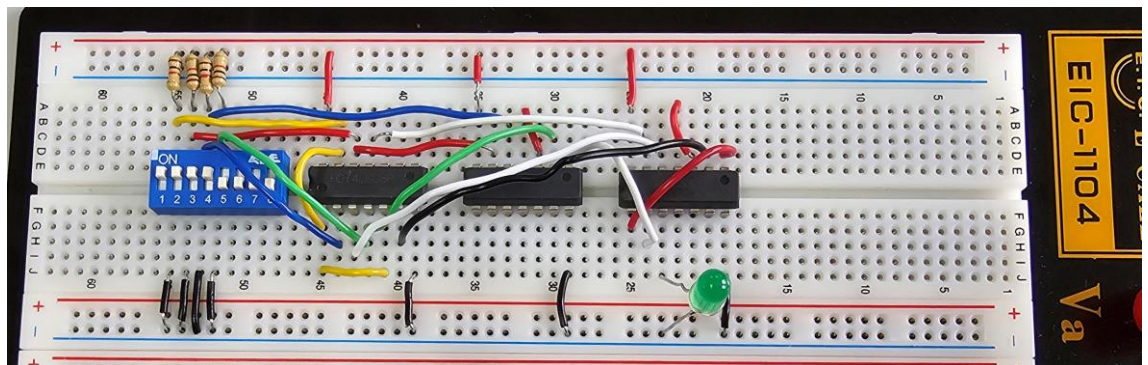
Karnaugh map

Simplest Boolean Expression: TA + TB + TC + ABC



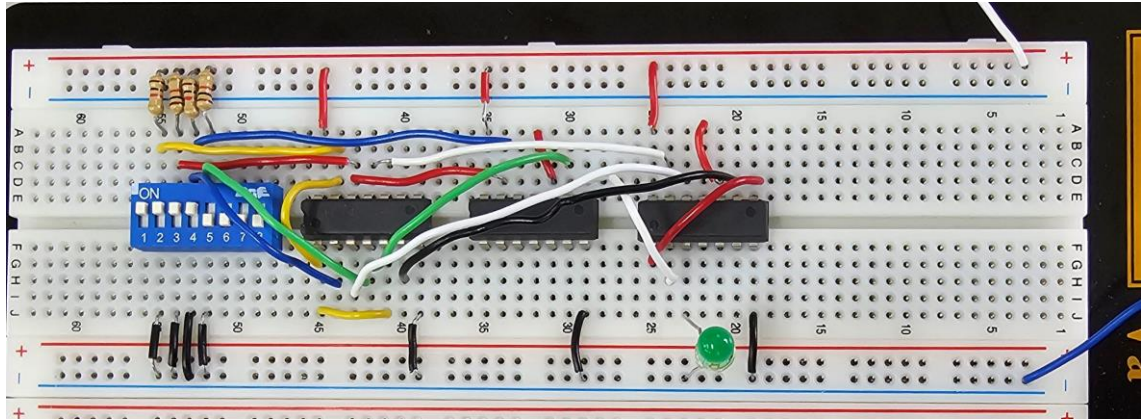
Breadboard Wiring Photos

Breadboard Wiring Photos (Please neatly connect the solid-core wires to the breadboard)

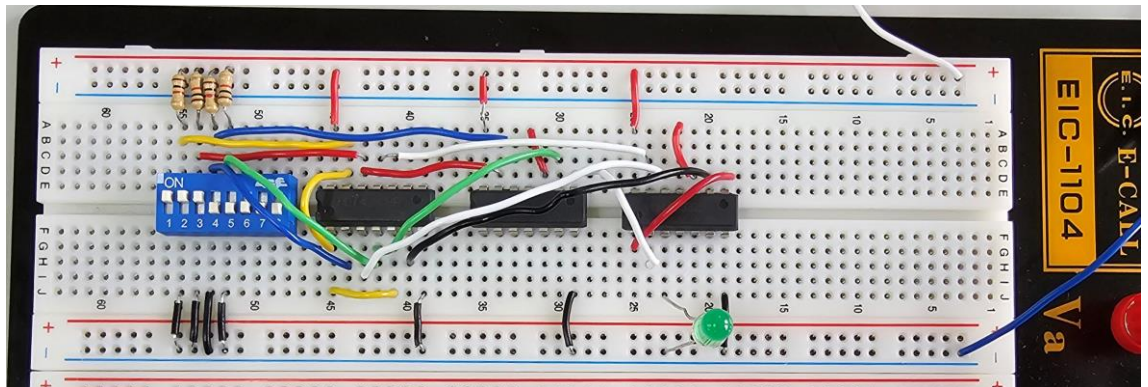


Experimental Results

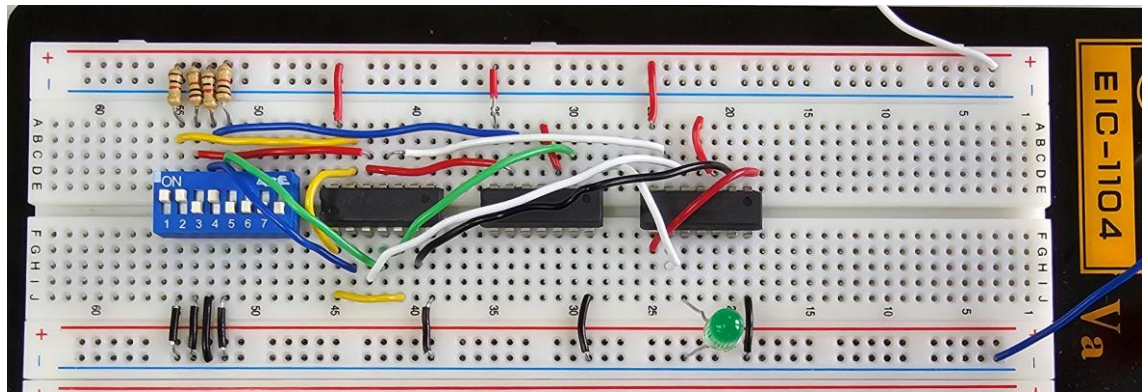
$(T=0, A=0, B=0, C=0)$ - The output must be connected to an LED.



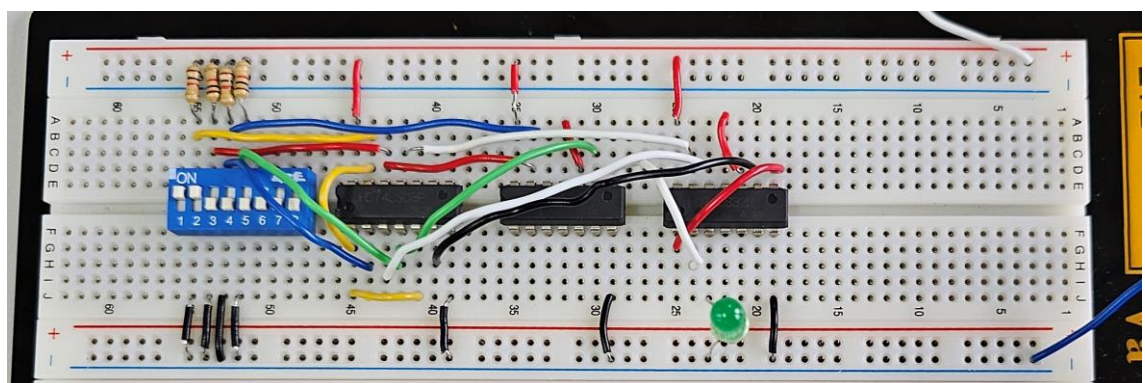
$(T=0, A=0, B=0, C=1)$ - The output must be connected to an LED.



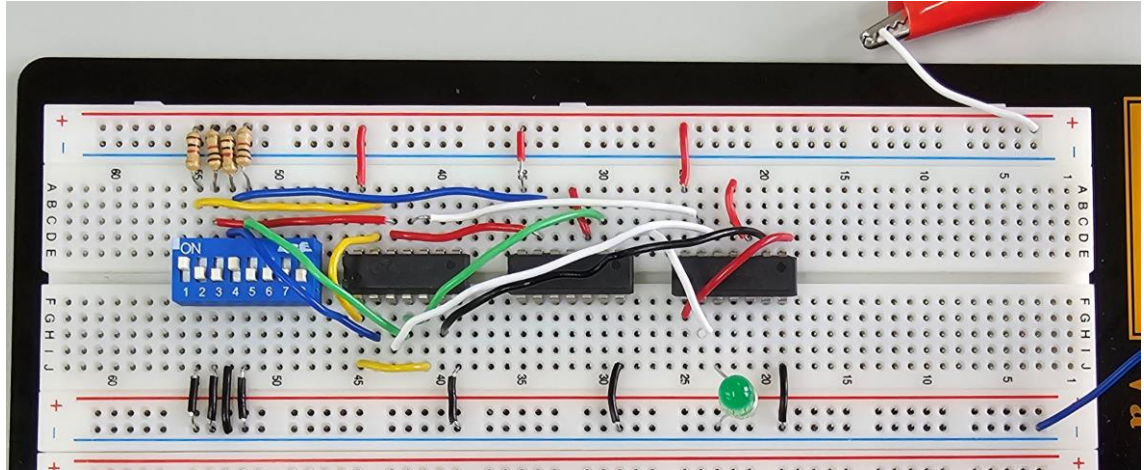
$(T=0, A=0, B=1, C=0)$ - The output must be connected to an LED.



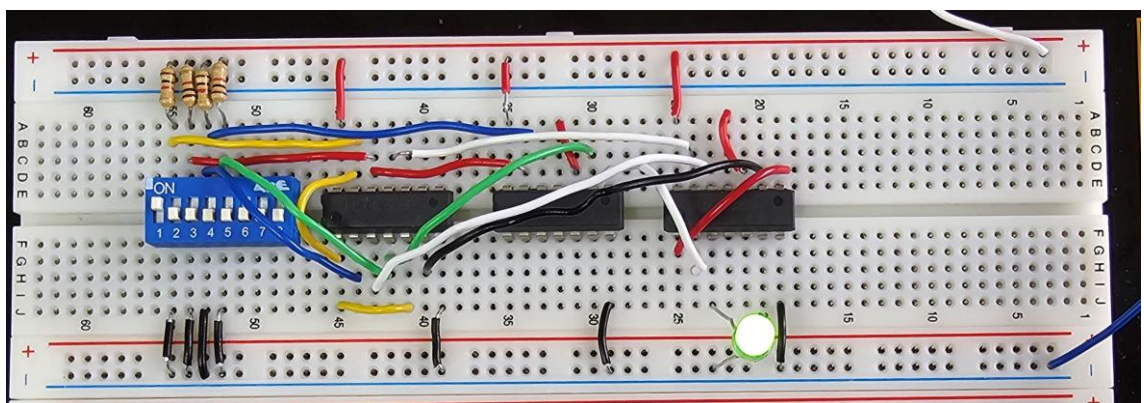
$(T=0, A=0, B=1, C=1)$ - The output must be connected to an LED.



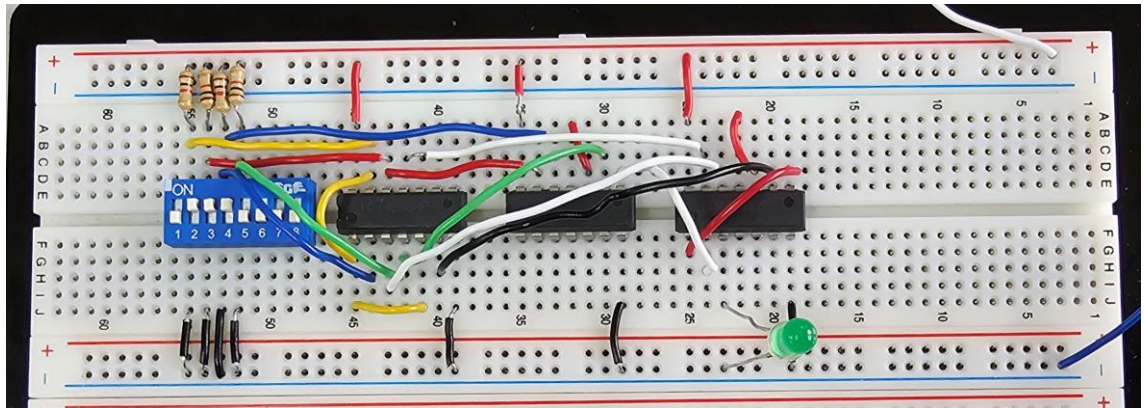
$(T=0, A=1, B=1, C=0)$ - The output must be connected to an LED.



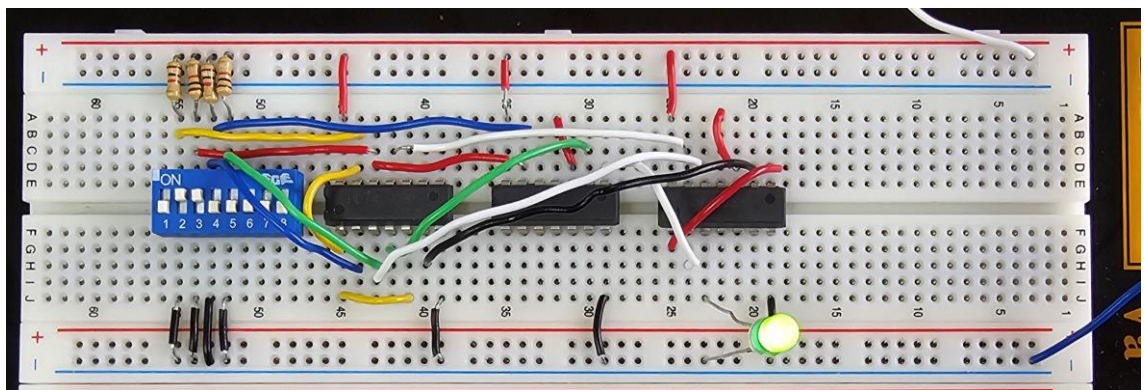
$(T=0, A=1, B=1, C=1)$ - The output must be connected to an LED.



$(T=1, A=0, B=0, C=0)$ - The output must be connected to an LED.

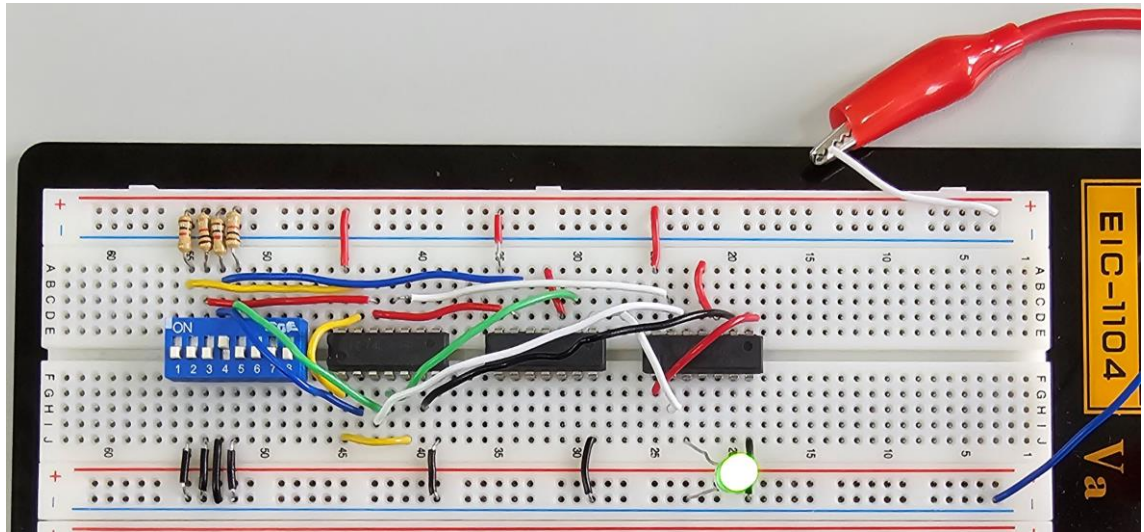


$(T=1, A=0, B=0, C=1)$ - The output must be connected to an LED.

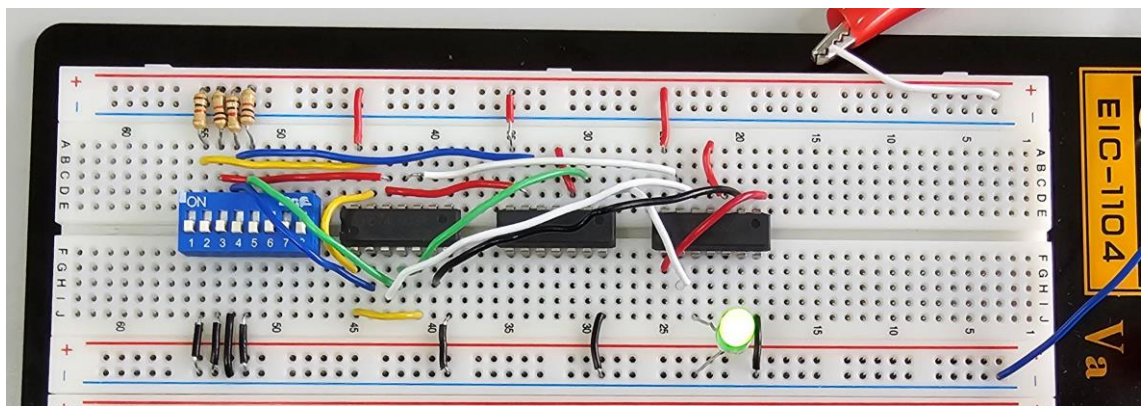


[illegible]

$(T=1, A=1, B=1, C=0)$ - The output must be connected to an LED.



$(T=1, A=1, B=1, C=1)$ - The output must be connected to an LED.



☆ Lab 3-2

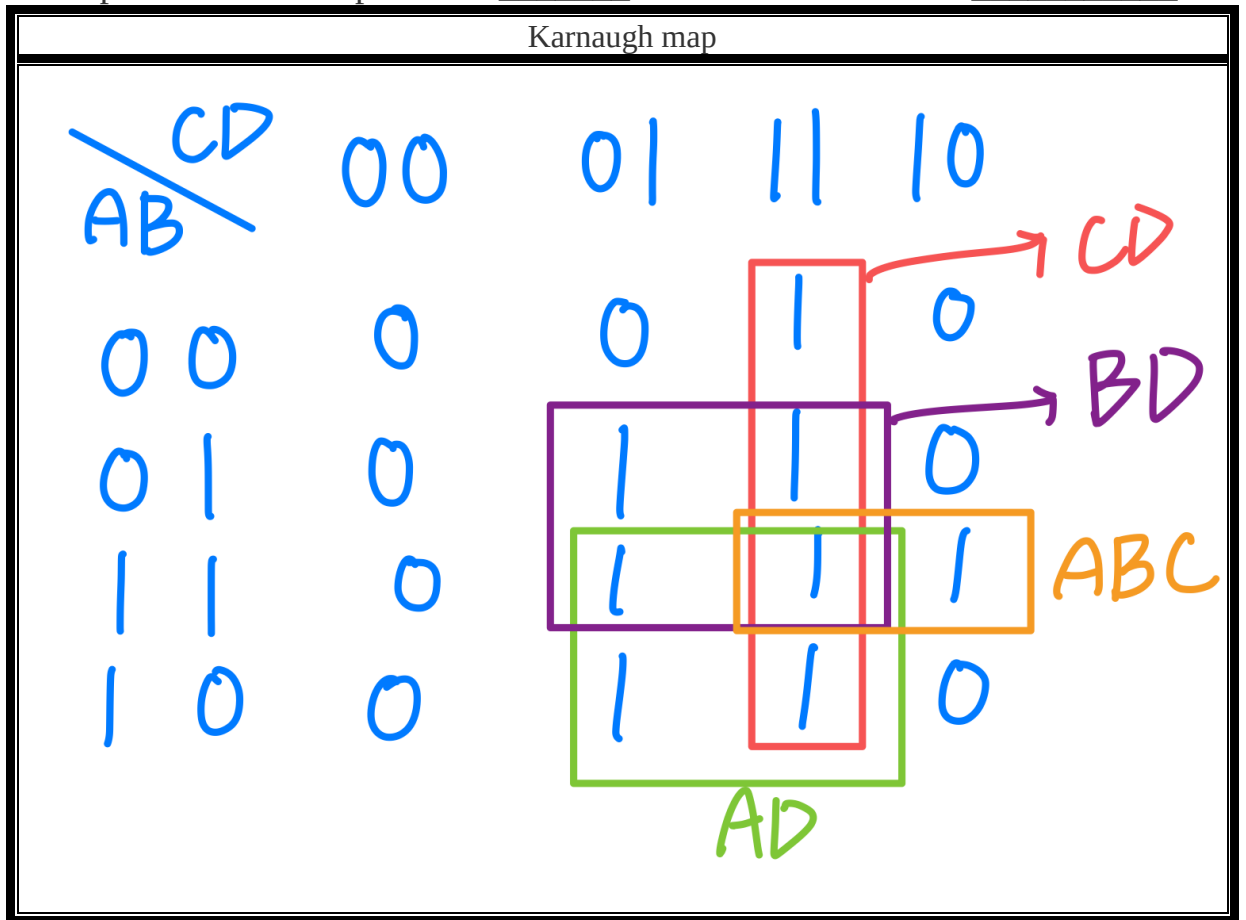
Truth Table (Logical)				
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Truth Table (Voltage)				
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
0	0	0	0	0.09
0	0	0	1	0.09
0	0	1	0	2.26
0	0	1	1	0.09
0	1	0	0	0.09
0	1	0	1	2.26
0	1	1	0	0.09
0	1	1	1	2.26
1	0	0	0	0.09
1	0	0	1	2.26
1	0	1	0	0.09
1	0	1	1	2.26
1	1	0	0	0.09
1	1	0	1	2.26
1	1	1	0	2.26
1	1	1	1	2.26



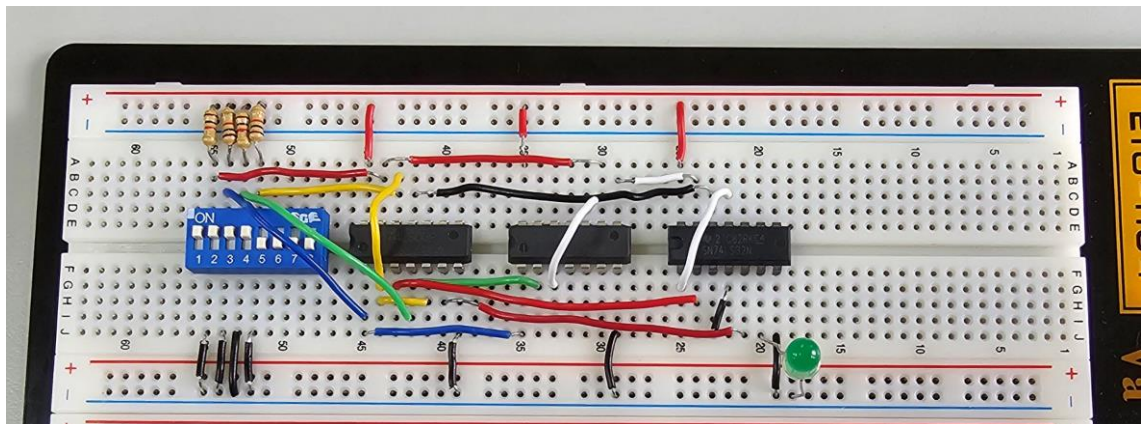
Karnaugh map

MSimplest Boolean Expression: $AD + CD + BD + ABC$



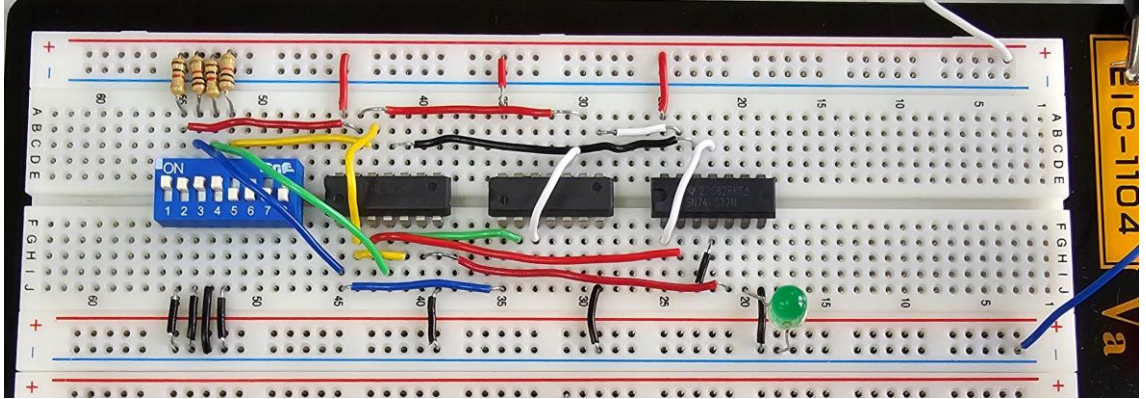
Breadboard Wiring Photos

Breadboard Wiring Photos (Please neatly connect the solid-core wires to the breadboard)

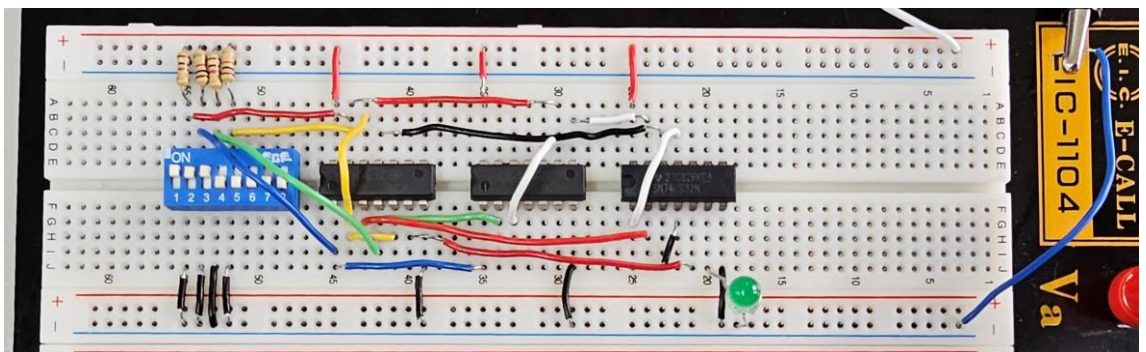


Experimental Results

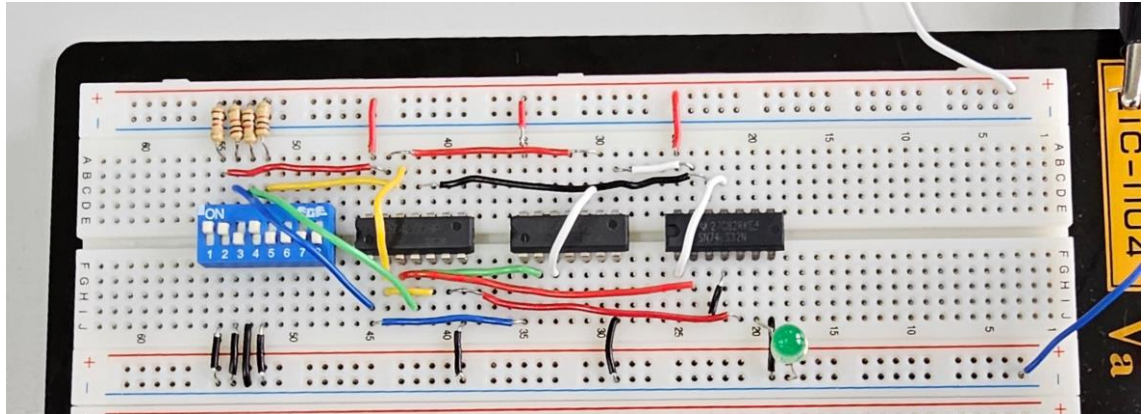
($A=0, B=0, C=0, D=0$) - The output must be connected to an LED.



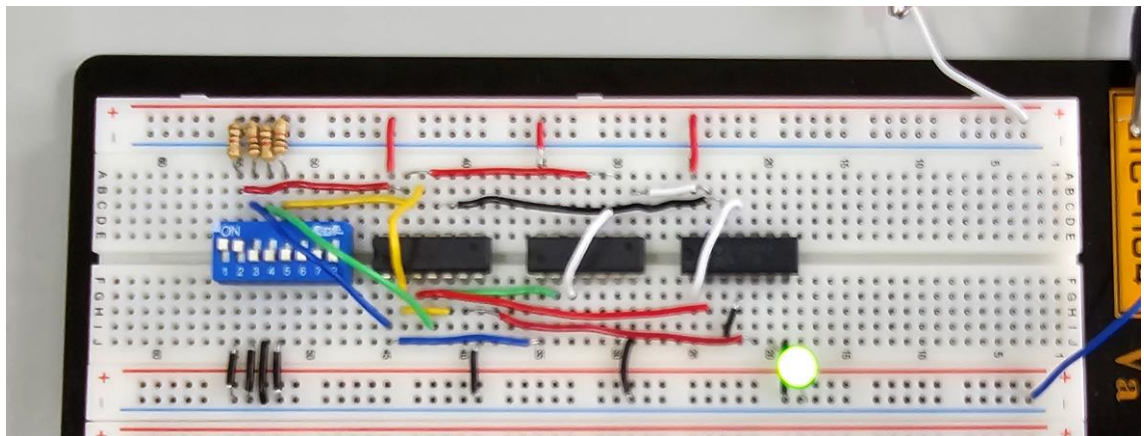
($A=0, B=0, C=0, D=1$) - The output must be connected to an LED.



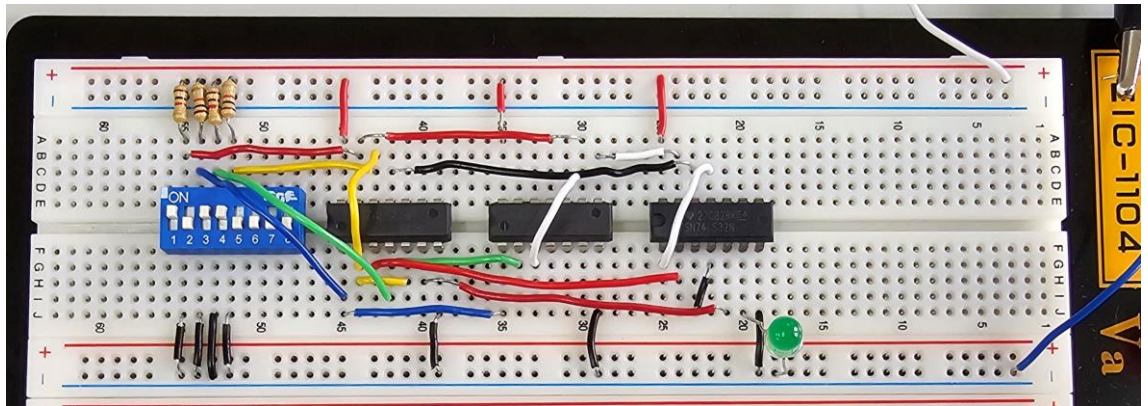
($A=0, B=0, C=1, D=0$) - The output must be connected to an LED.



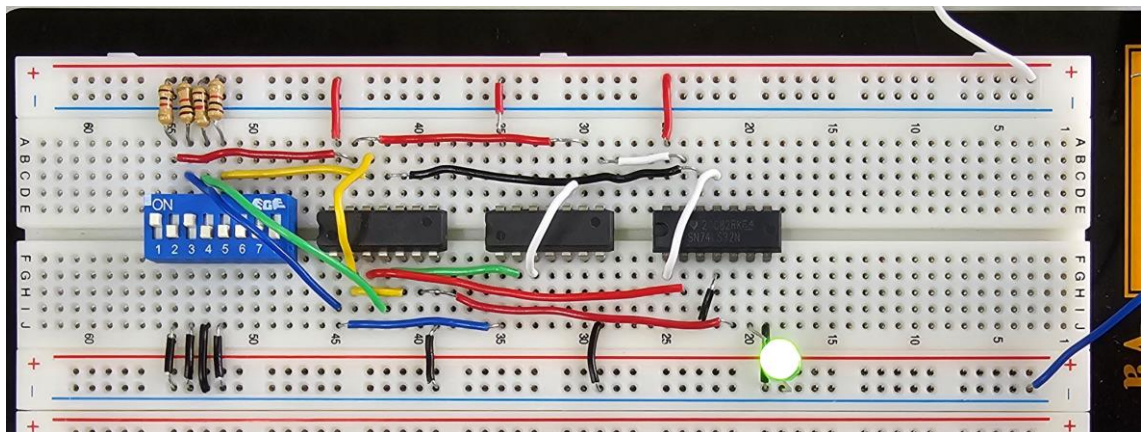
($A=0, B=0, C=1, D=1$) - The output must be connected to an LED.



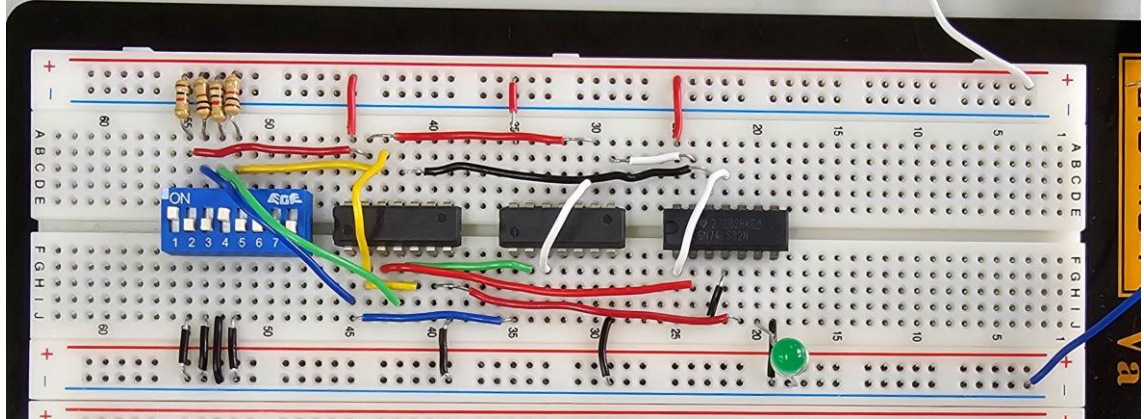
($A=0, B=1, C=0, D=0$) - The output must be connected to an LED.



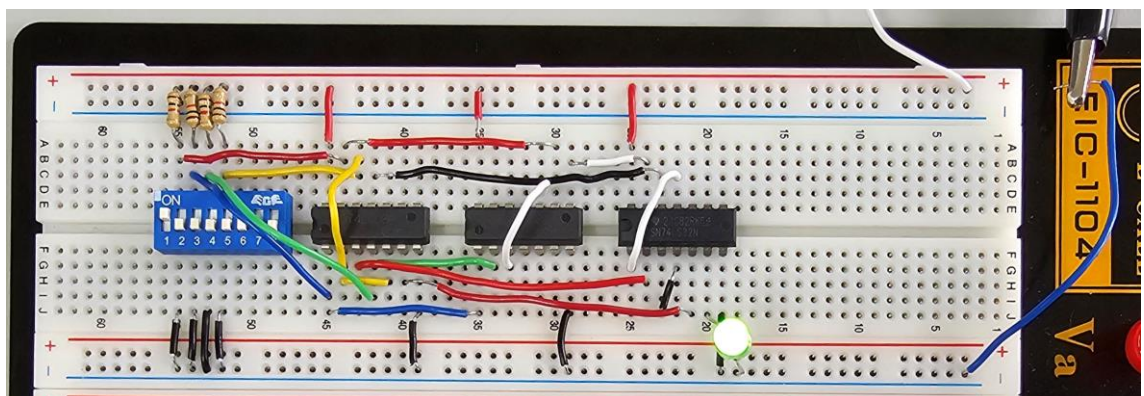
($A=0, B=1, C=0, D=1$) - The output must be connected to an LED.



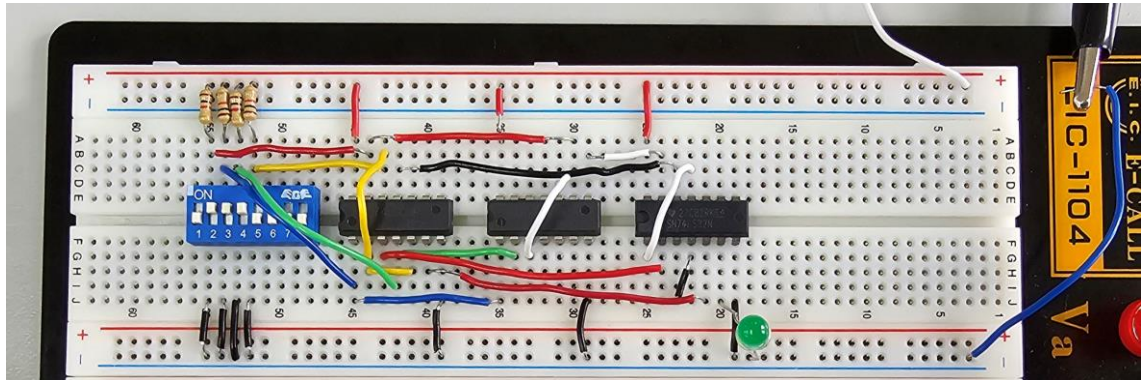
(A=0, B=1, C=1, D=0) - The output must be connected to an LED.



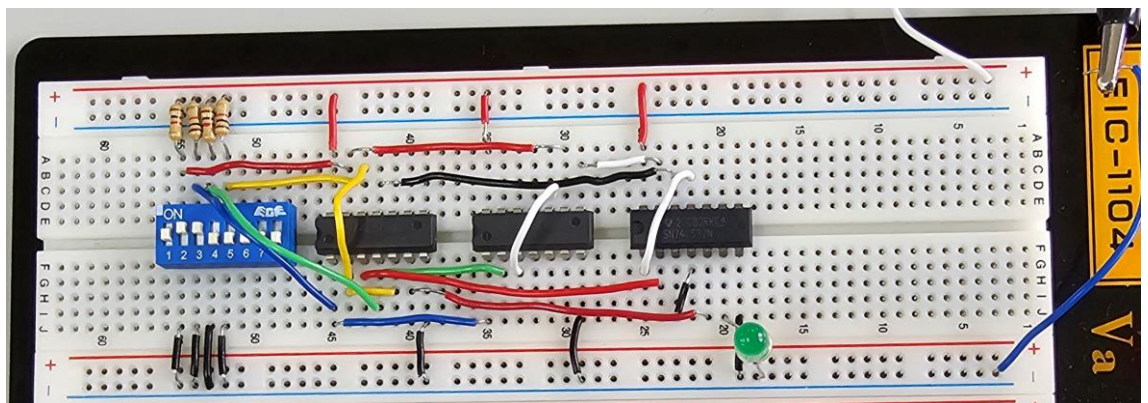
(A=0, B=1, C=1, D=1) - The output must be connected to an LED.



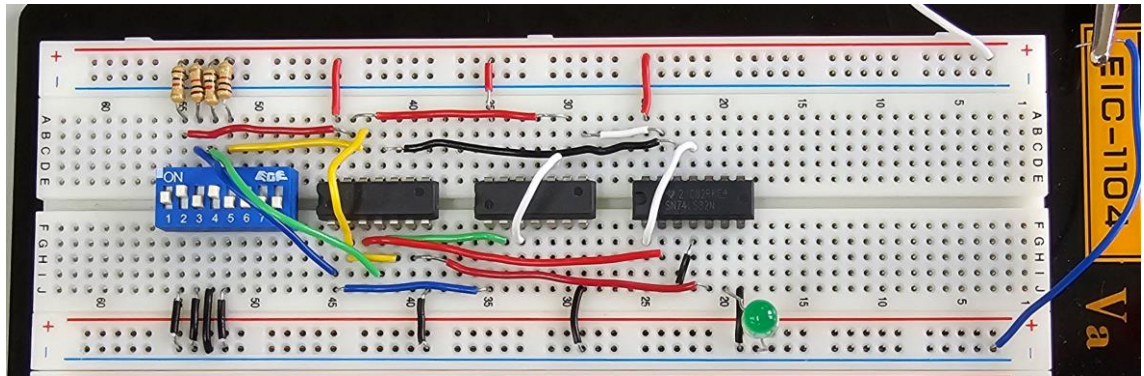
($A=1, B=0, C=0, D=0$) - The output must be connected to an LED.



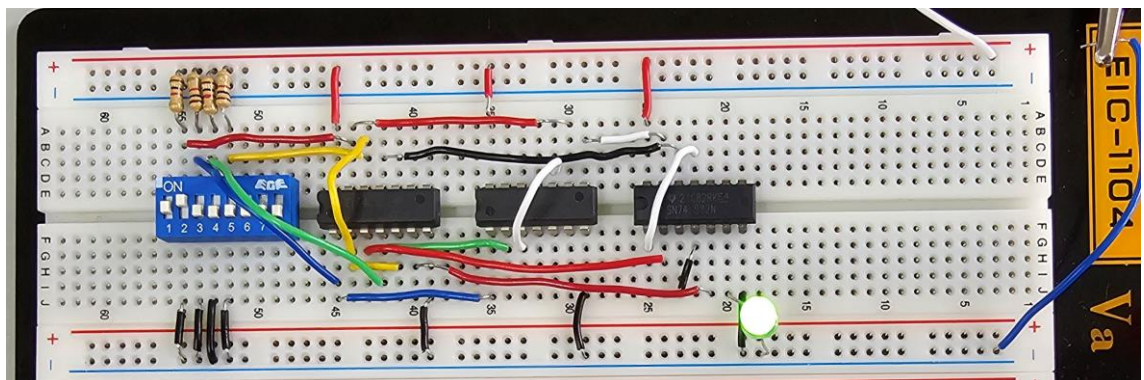
($A=1, B=0, C=0, D=1$) - The output must be connected to an LED.



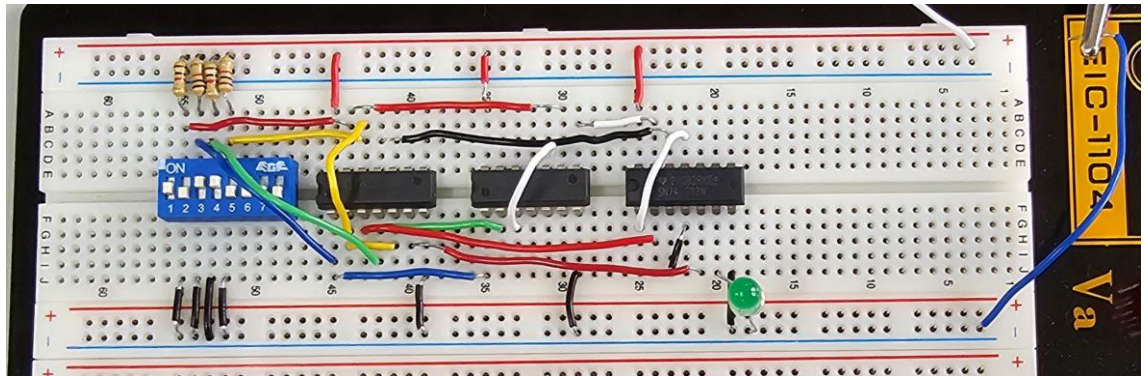
($A=1, B=0, C=1, D=0$) - The output must be connected to an LED.



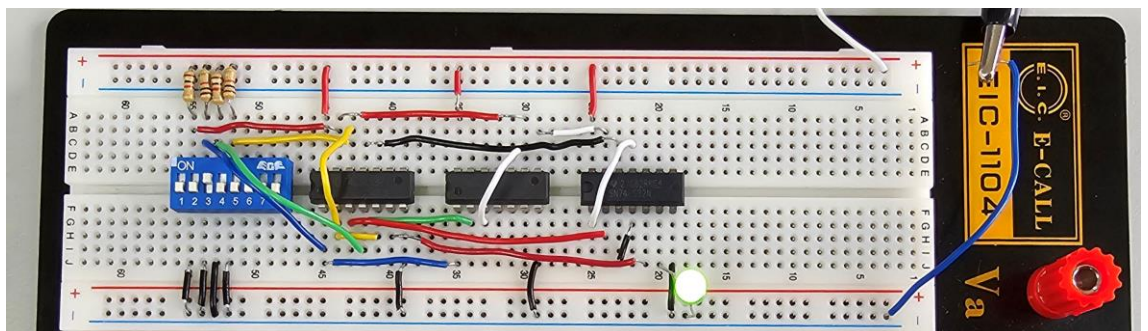
($A=1, B=0, C=1, D=1$) - The output must be connected to an LED.



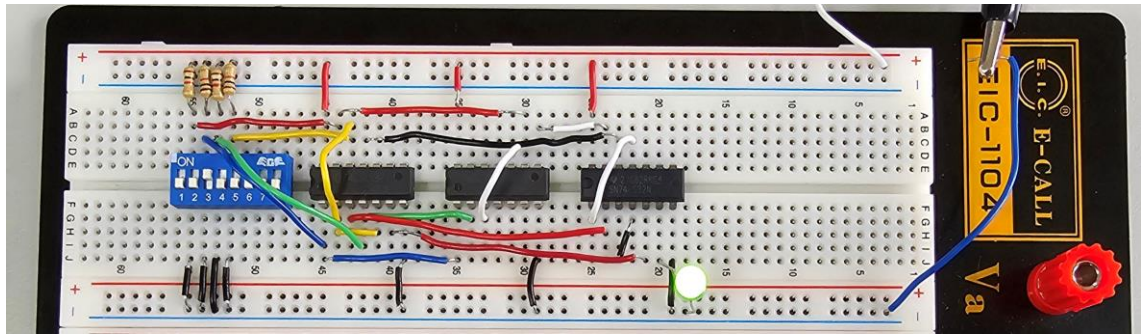
(A=1, B=1, C=0, D=0) - The output must be connected to an LED.



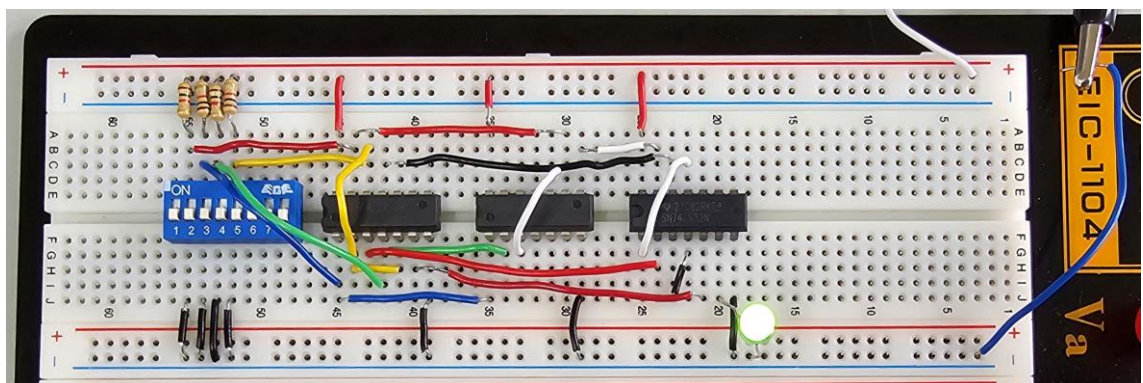
(A=1, B=1, C=0, D=1) - The output must be connected to an LED.



(A=1, B=1, C=1, D=0) - The output must be connected to an LED.



(A=1, B=1, C=1, D=1) - The output must be connected to an LED.



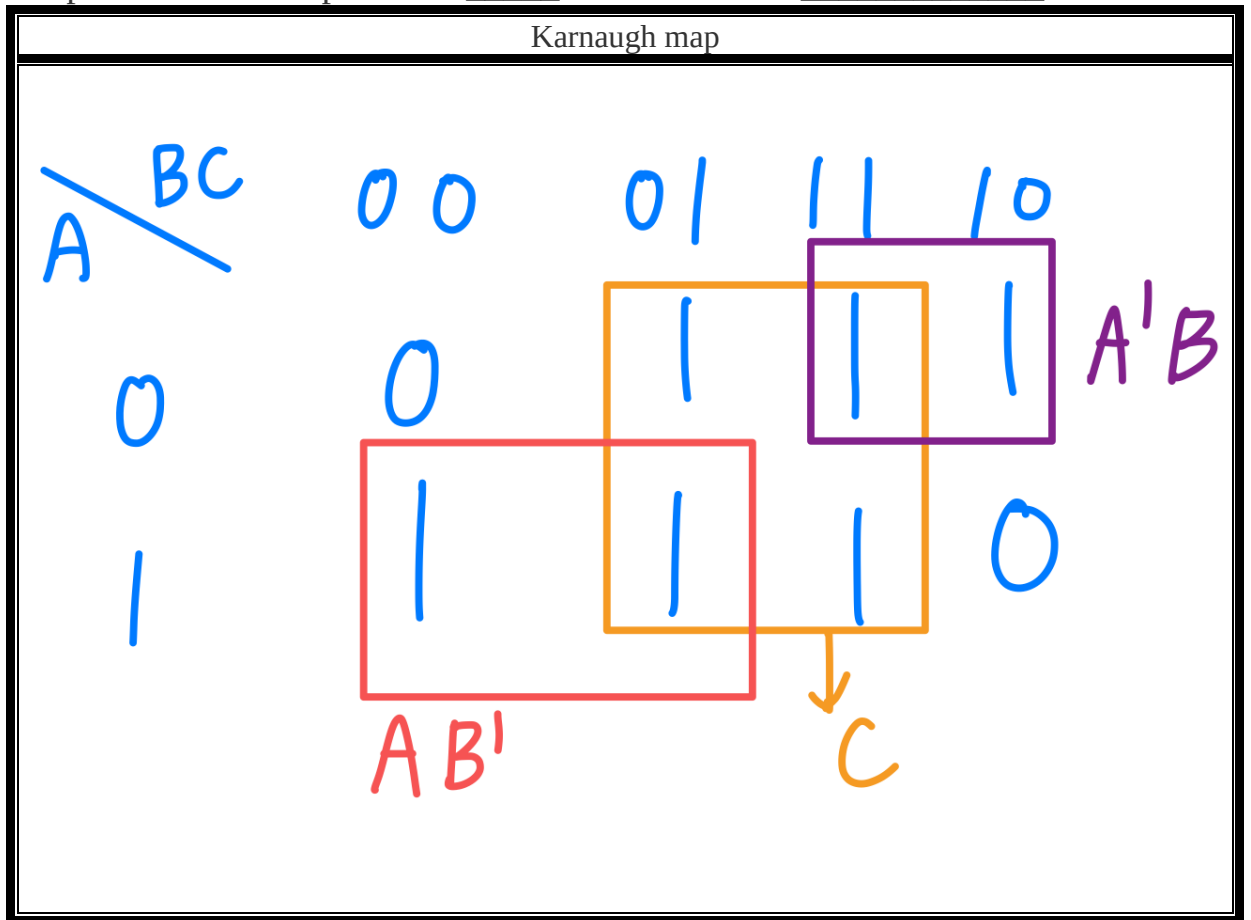
☆ Lab 3-3

Truth Table (Logical)			
<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1
Truth Table (Voltage)			
<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	0.09
0	0	1	2.26
0	1	0	2.26
0	1	1	2.26
1	0	0	2.26
1	0	1	2.26
1	1	0	0.09
1	1	1	2.26



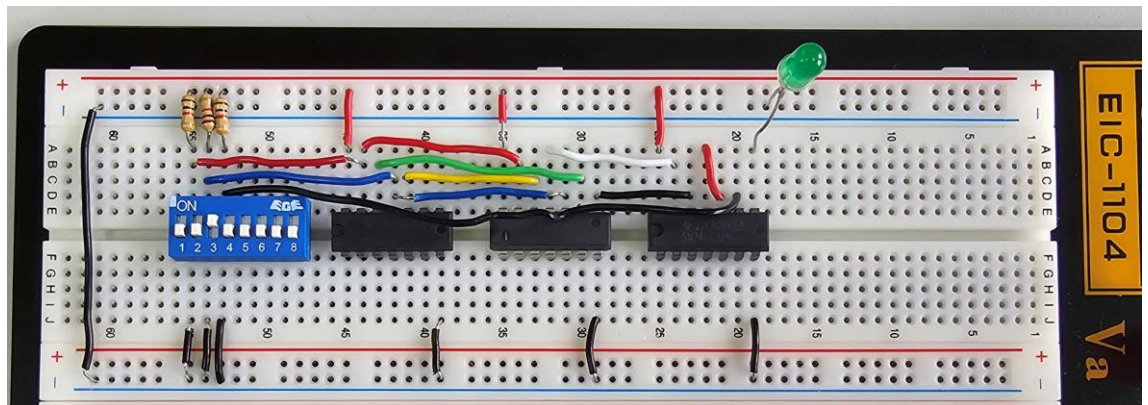
Karnaugh map

Simplest Boolean Expression: $AB' + A'B + C$



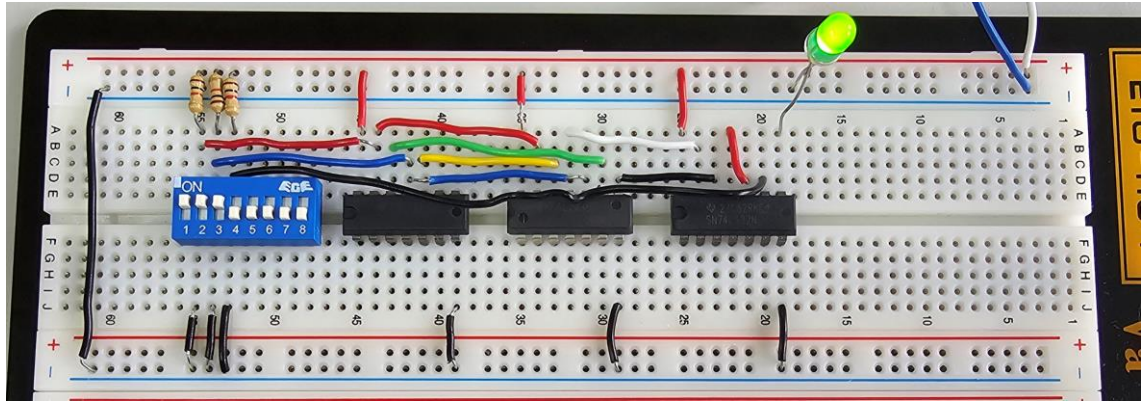
Breadboard wiring Photos

Breadboard Wiring Photos (Please neatly connect the solid-core wires to the breadboard)

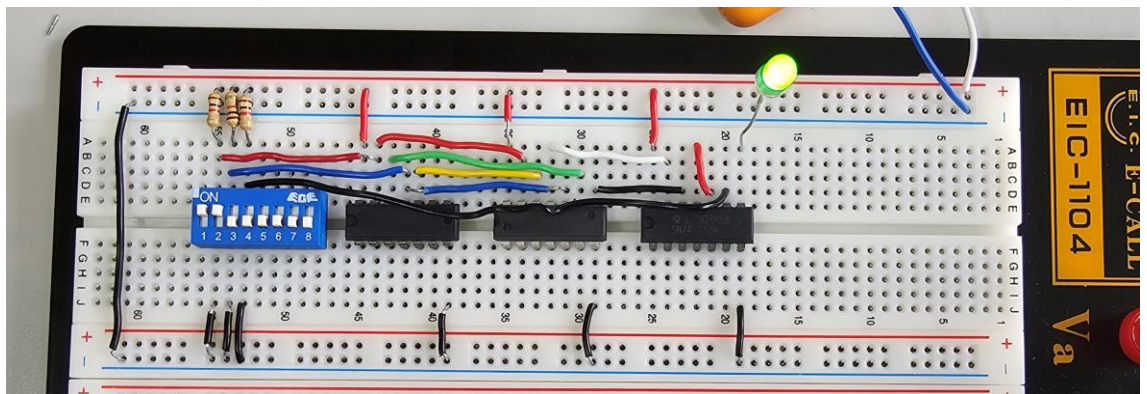


Experimental results

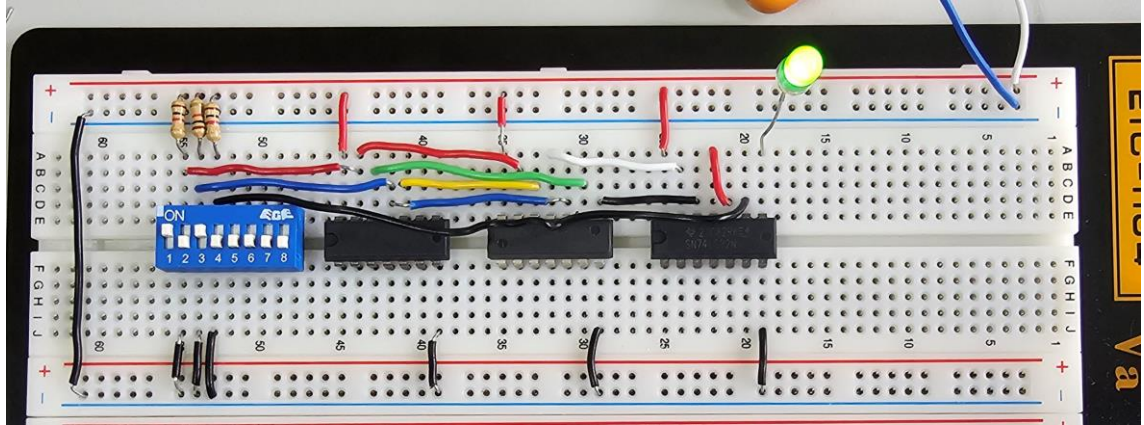
(A=0, B=0, C=0) - The output must be connected to an LED.



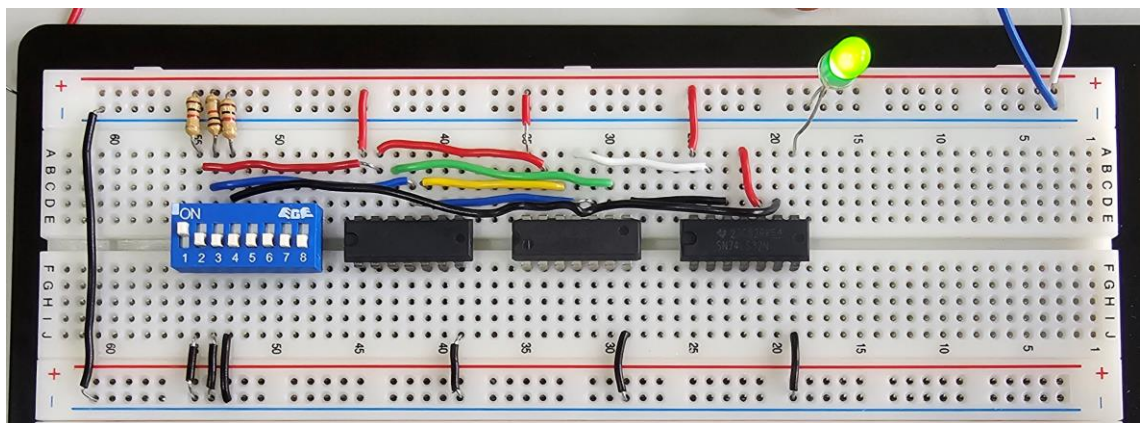
(A=0, B=0, C=1) - The output must be connected to an LED.



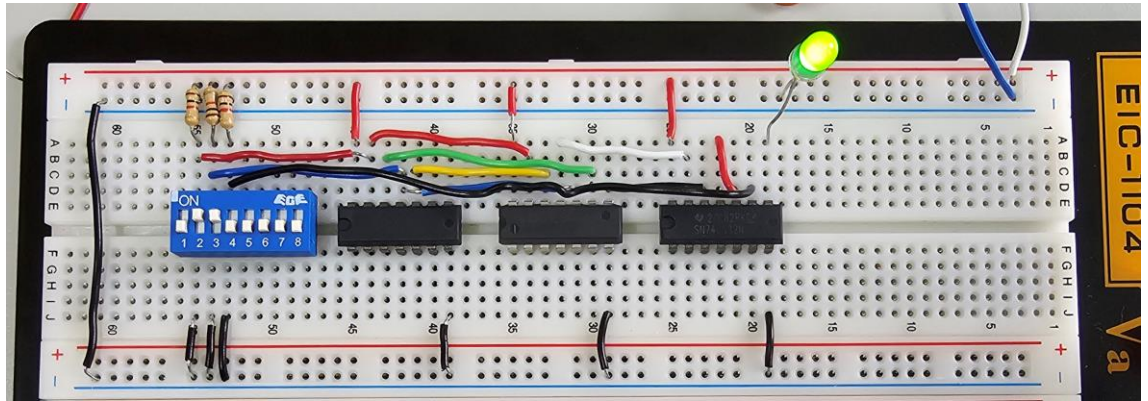
(A=0, B=1, C=0) - The output must be connected to an LED.



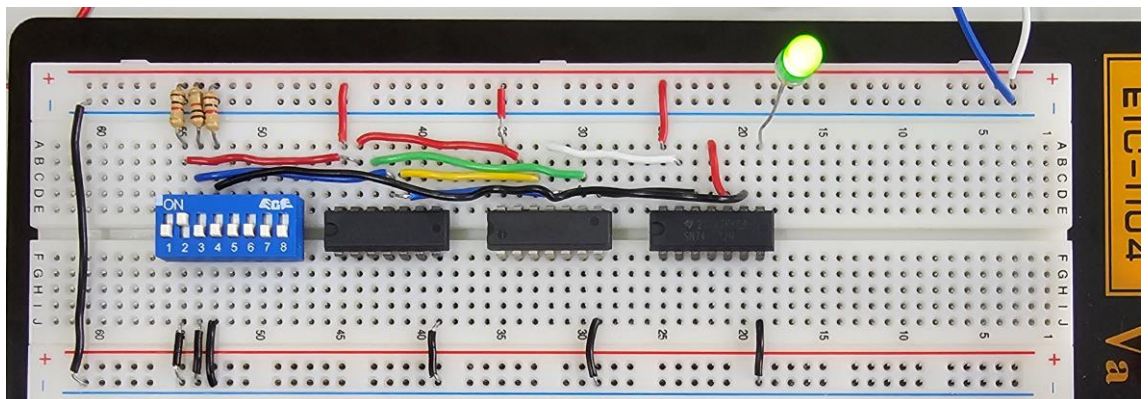
(A=0, B=1, C=1) - The output must be connected to an LED.



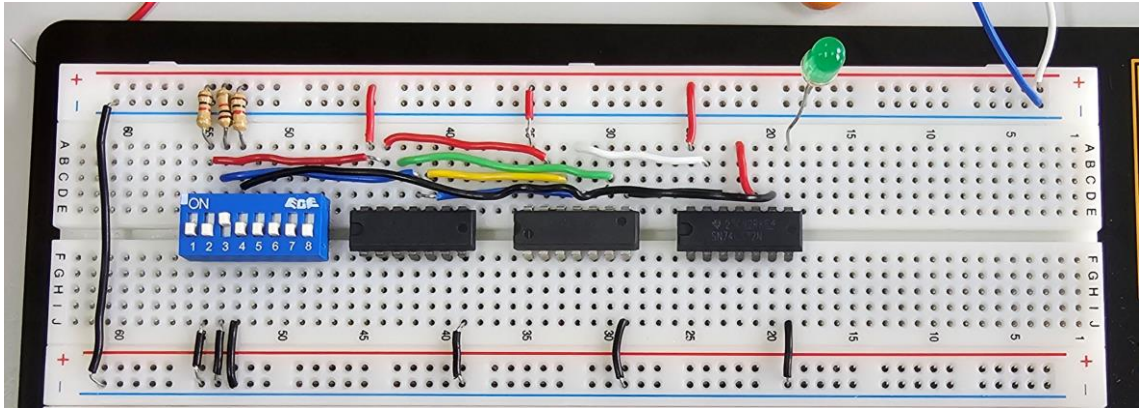
($A=1, B=0, C=0$) - The output must be connected to an LED.



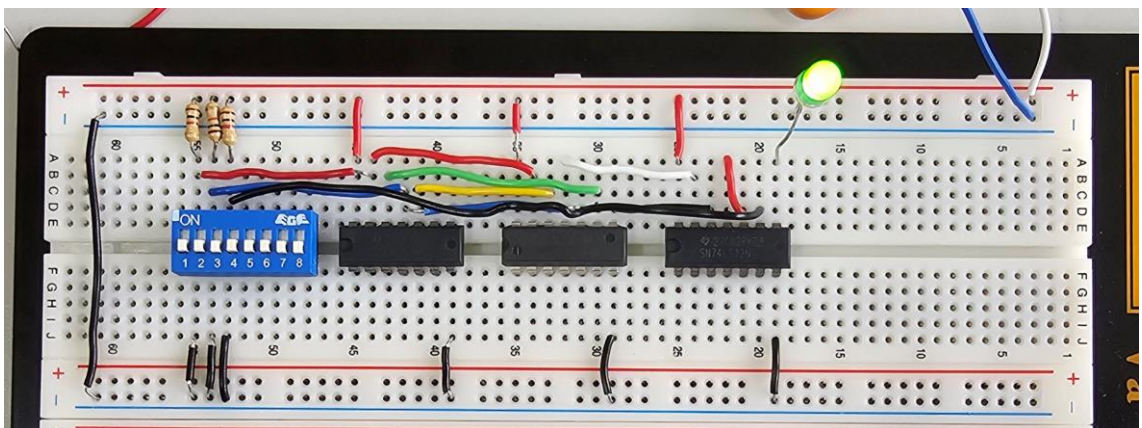
($A=1, B=0, C=1$) - The output must be connected to an LED.



(A=1, B=1, C=0) - The output must be connected to an LED.



(A=1, B=1, C=1) - The output must be connected to an LED.



Lab Summary and Reflection :

Summary

In this week's Digital Logic Lab, we focused on implementing basic logic gate circuits using TTL ICs such as 7400, 7404, 7408, and 7432. The experiment involved constructing truth tables, simplifying Boolean expressions using techniques like K-map and Boolean algebra, and then verifying our results by wiring the circuits on a breadboard with LED outputs. We divided the lab into several parts, each addressing different logic functions and ensuring that both the theoretical and practical aspects of digital circuit design were thoroughly integrated.

Reflection

Reflecting on this lab, I appreciated how the hands-on approach reinforced the fundamental concepts we learned in class. A significant improvement from previous sessions was the resolution of the hardware issue; earlier reports highlighted a shortage of IC chips that impeded circuit replacement, but in today's session, we had ample components available. This allowed for smoother troubleshooting and experimentation. Moreover, the requirement to derive the simplest Boolean expression prior to wiring challenged us to revisit and apply our knowledge of K-maps and Boolean simplification, ultimately enhancing our understanding of digital logic and our ability to implement practical solutions effectively.